# AGPO: Anchor-Guided Progressive Optimization for Enhancing 3D Gaussian Representations

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**Abstract**

Recent advances in 3D generation and reconstruction enable the efficient construction of 3D Gaussian representations from multi-view observations. Nevertheless, imperfect observations and reconstruction errors often leave degraded details, geometric artifacts, and unstable cross-view appearance in the resulting representations. We propose Anchor-Guided Progressive Optimization (AGPO), a lightweight, training-free framework for enhancing initialized 3D Gaussian representations. Rather than treating supervision from different viewpoints as isolated constraints, AGPO organizes the available observations as anchor-guided supervision and exploits inter-view geometry to extend reliable structural and appearance information toward under-constrained regions. Depth-guided reprojection establishes explicit correspondences between neighboring viewpoints, while deterministic wavefront propagation carries the resulting supervision across the view space. Throughout propagation, an anchor retention constraint preserves fidelity at directly observed viewpoints. Meanwhile, an Uncertainty-Aware Reweighting (UAR) mechanism adaptively adjusts pixel-level supervision based on depth consistency, effectively suppressing artifacts in occluded or uncertain regions.

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Experiments in both text-conditioned 3D generation and multi-view reconstruction settings demonstrate the effectiveness of AGPO. On the $T^3$Bench Single Object benchmark, AGPO increases the $T^3$Bench Overall score of the initial representation from 46.84 to 52.68 and reduces the reprojection mean squared error from $0.929 \times 10^{-3}$ to $0.736 \times 10^{-3}$. Consistent improvements on FreeSplatter-initialized representations further demonstrate its generalization across different initialization strategies. Project page: `https://cola012.github.io/AGPO/`

*Keywords:* 3D Gaussian Splatting; 3D Representation Enhancement; Multi-view Consistency; Post-optimization; Sparse-view Reconstruction.

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## 1. Introduction

3D content generation has advanced rapidly in recent years, enabling the efficient creation of digital assets with plausible geometry and appearance. In particular, text-conditioned 3D generation allows users to synthesize semantically meaningful content directly from natural language descriptions [1, 2], substantially reducing the expertise and manual effort required by conventional 3D modeling. These advances support applications including e-commerce [3], virtual reality [4], gaming [5], film production [6], and digital content creation [7]. Nevertheless, efficient generation does not necessarily guarantee high-quality 3D representations. Generated results may still contain degraded details, geometric artifacts, and view-dependent inconsistencies, especially at viewpoints that are insufficiently constrained by the available observations. Improving the visual fidelity, geometric plausibility, and cross-view stability of generated 3D representations therefore remains an important problem.

Recent 3D generation pipelines increasingly adopt 3D Gaussian Splatting (3DGS) [8] because of its explicit representation, differentiable formulation, and efficient rendering. Multi-view generative models such as MVDream [9] can first synthesize observations of an object from different viewpoints, which are subsequently lifted into a 3D representation through multi-view-to-3D reconstruction [10]. Although this paradigm

substantially improves generation efficiency, subtle inconsistencies among the generated views, limited view coverage, and reconstruction inaccuracies may be inherited by the resulting 3DGS. Consequently, the initialized representation may exhibit degraded appearance, incomplete or spurious geometry, and unstable renderings at intermediate or novel viewpoints.

Existing methods alleviate these problems using stronger 3D priors, multi-view diffusion guidance, depth-aware constraints, perceptual objectives, and coarse-to-fine refinement strategies [11, 12]. However, many of these techniques primarily improve the generation process rather than directly refining the resulting 3D representation. Moreover, post-optimization commonly applies supervision to individual or independently sampled viewpoints without fully exploiting the geometric dependencies among neighboring views. Improvements at directly supervised viewpoints may therefore remain localized, while unreliable correspondences caused by occlusion or inaccurate geometry may introduce erroneous constraints. Effective refinement should instead organize supervision according to inter-view geometry, propagate reliable information toward under-constrained viewpoints, and suppress unreliable propagation.

Recent work has also explored progressive refinement of existing 3D representations. DIFIX3D+ [13] renders interpolated novel views, corrects their artifacts using a learned diffusion-based image enhancer, and distills the refined pseudo-observations back into the 3D representation. Although effective, this approach requires an additional learned generative model to construct refined supervision. Meanwhile, available multi-view observations and the initialized representation already contain useful appearance and structural information. This motivates us to investigate whether these signals can be exploited directly without introducing an additional generative enhancement model.

Based on this motivation, we propose Anchor-Guided Progressive Optimization (AGPO), a lightweight and training-free post-optimization framework for enhancing

initialized 3D Gaussian representations. AGPO uses available multi-view observations as anchor references and organizes their appearance and structural information into progressive cross-view supervision. Depth-guided reprojection establishes correspondences between neighboring viewpoints, while deterministic wavefront propagation extends reliable supervision from anchor regions toward under-constrained regions of the view space. An anchor retention constraint preserves fidelity to directly observed views throughout propagation. Furthermore, Uncertainty-Aware Reweighting (UAR) estimates pixel-level supervision reliability from depth consistency and suppresses uncertain correspondences caused by occlusion or inaccurate geometry. Together, these components improve appearance quality and cross-view consistency without retraining or modifying the underlying generation or reconstruction model.

Extensive experiments under both 3D generation and multi-view reconstruction settings demonstrate the effectiveness of AGPO. On the $T^3$Bench Single Object benchmark, AGPO improves the $T^3$Bench Overall score of the initial Large Multi-View Gaussian Model (LGM) [10] representation from 46.84 to 52.68 and reduces the reprojection mean squared error (MSE) from $0.929 \times 10^{-3}$ to $0.736 \times 10^{-3}$. Consistent improvements on FreeSplatter-initialized representations further demonstrate the generalization of AGPO across different initialization strategies.

The main contributions of this paper are summarized as follows:

- We introduce AGPO, a lightweight and training-free post-optimization framework that directly enhances initialized 3D Gaussian representations using available multi-view observations, without retraining or modifying the underlying generation or reconstruction model.
- We develop an anchor-guided progressive optimization strategy that exploits inter-view geometry to extend appearance and structural constraints from directly supervised viewpoints toward under-constrained regions while preserving fidelity to the anchor observations.

- We propose a UAR mechanism that estimates pixel-level supervision reliability from depth consistency and suppresses unreliable correspondences in occluded or geometrically ambiguous regions.

The remainder of this paper is organized as follows. Section 2 reviews related work on 3D generation, 3D representation, and 3D representation enhancement. Section 3 presents the overall AGPO framework and details the anchor-guided progressive optimization and uncertainty-aware reweighting mechanisms. Section 4 describes the experimental settings and reports the ablation studies, quantitative and qualitative comparisons, user study, and discussion. Finally, Section 5 concludes the paper.

## 2. Related Work

Recent advances in diffusion-based generation and efficient 3D representations have substantially improved the quality and efficiency of 3D content creation. In this section, we review related studies on 3D generation, 3D representations, and post-generation or post-reconstruction enhancement.

### *2.1. 3D Generation Methods*

Diffusion-based text-to-3D methods commonly transfer pretrained 2D generative priors to 3D representations. DreamFusion [14] introduces Score Distillation Sampling (SDS) to optimize a 3D representation from rendered views under the guidance of a text-conditioned diffusion model. Subsequent methods, including Magic3D [15], ProlificDreamer [16], and Consistent3D [12], further improve visual fidelity, geometric plausibility, and cross-view consistency through coarse-to-fine optimization, improved score distillation, and geometric constraints [17, 18]. Nevertheless, these optimization-based approaches generally require costly per-instance optimization and may still exhibit unstable geometry or appearance across viewpoints.

To improve generation efficiency, recent methods increasingly adopt feed-forward or direct generation paradigms. View-aware generative models such as MVDream [9]

first synthesize multi-view observations, which can then be lifted into a 3D representation by feed-forward reconstruction methods such as LGM [10] and Instant3D [19]. Other approaches encode 3D structures into image-like or latent representations and adapt pretrained diffusion models for direct 3D generation [20, 21, 22]. Although these methods produce increasingly strong initial representations, inconsistent observations, limited view coverage, and reconstruction errors may still leave appearance or geometric artifacts, motivating lightweight refinement of the resulting 3D representations.

### *2.2. 3D Representation Methods*

Neural Radiance Fields (NeRF) [23] model scene density and appearance using neural networks and achieve high-quality novel-view synthesis through volumetric rendering. However, the required ray marching and dense sampling often result in substantial optimization and rendering costs. More recently, 3DGS [8] has emerged as an efficient explicit representation that supports high-quality rendering and differentiable optimization through Gaussian primitives and fast rasterization.

The efficiency and flexibility of 3DGS have led to its broad adoption in both generation and reconstruction [24, 25, 26]. GaussianDreamer [27] combines 2D and 3D diffusion priors to initialize and optimize 3D Gaussians, whereas LGM [10] directly predicts a 3D Gaussian representation from sparse multi-view images. FreeSplatter [28] further reconstructs 3D Gaussians from uncalibrated sparse-view images while estimating the corresponding camera parameters [29]. Despite their efficiency, the resulting representations remain dependent on observation quality, view coverage, and reconstruction accuracy, and may contain degraded appearance, geometric artifacts, or unstable renderings in under-constrained regions [30, 31].

### *2.3. 3D Representation Enhancement*

Recent studies have explored post-generation and post-reconstruction enhancement to improve existing 3D representations. SuperGaussian [32] repurposes pretrained

video upsampling models to super-resolve renderings of low-resolution 3D representations and refits the enhanced observations into a higher-resolution 3D Gaussian representation. 3DGS-Enhancer [33] uses view-consistent video diffusion priors to enhance novel views rendered from sparsely reconstructed 3DGS models and subsequently refine the underlying representation. 3DEnhancer [34] introduces a multi-view latent diffusion model that enhances renderings from coarse 3D inputs while preserving cross-view consistency; the enhanced views can be reconstructed into a new representation or used as supervision for per-instance optimization.

DIFIX3D+ [13] employs a single-step diffusion model to remove artifacts from under-constrained novel views and distills the refined pseudo-observations back into NeRF or 3DGS representations. FreeFix [35] avoids diffusion-model fine-tuning and performs interleaved 2D–3D refinement using a pretrained image diffusion model with pixel-level confidence guidance. Photo3D [36] improves the photorealism of 3D-native generation by constructing structure-aligned and detail-enhanced multi-view supervision for different 3D generative paradigms.

These methods demonstrate the effectiveness of generative priors for enhancing appearance and under-constrained regions. However, most require an additional learned or pretrained generative model to produce refined supervision, and some further require dedicated enhancement data or model adaptation [37, 38]. In contrast, AGPO directly exploits available multi-view observations and their geometric relationships. It progressively propagates existing supervision across the view space while accounting for correspondence reliability, providing a training-free post-optimization strategy without an additional generative refinement model.

## 3. Our Approach

In this section, we first present the overall framework of AGPO (Section 3.1). We then detail the anchor-guided progressive optimization strategy, which propagates

structural and appearance constraints from anchor references to neighboring viewpoints (Section 3.2). Finally, we introduce UAR, which suppresses unreliable supervision in occluded or geometrically ambiguous regions based on depth consistency (Section 3.3). An overview of the proposed framework is illustrated in Fig. 1.

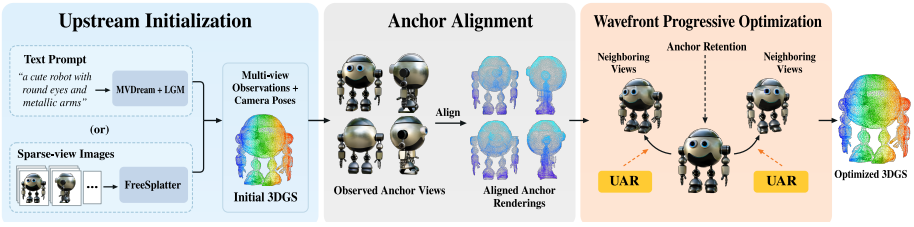



Figure 1: **Overview of the proposed AGPO framework.** AGPO takes multi-view observations, camera poses, and an initial 3DGS representation obtained from either a text-to-3D pipeline (MVDream and LGM) or sparse-view reconstruction (FreeSplatter). It first aligns the initial representation with the observed anchor views and then progressively propagates reliable structural and appearance constraints to neighboring viewpoints. Anchor retention preserves reliable anchor supervision, while uncertainty-aware reweighting (UAR) suppresses unreliable cross-view correspondences during propagation, producing a more geometrically coherent and cross-view-consistent 3DGS representation.

### *3.1. Overview Framework*

AGPO is a training-free post-optimization framework designed to enhance the visual fidelity and cross-view consistency of 3D Gaussian representations. As illustrated in Fig. 1, given a set of multi-view images and their corresponding camera poses, an initial 3D Gaussian representation is first constructed through a multi-view-to-3D reconstruction pipeline. These multi-view images serve as anchor references and provide supervision signals for subsequent optimization.

AGPO first aligns the initialized Gaussian representation with the anchor references to establish reliable supervision signals. Based on the aligned anchors, AGPO performs progressive optimization over neighboring viewpoints by propagating structural and appearance constraints along the camera trajectory. To improve the robustness of propagation, uncertainty-aware reweighting adaptively adjusts pixel-level supervision according to geometric reliability. Meanwhile, an anchor retention constraint preserves the fidelity of the anchor views during optimization.

Through the integration of anchor-based initialization, progressive propagation, reliability-aware supervision, and anchor retention, AGPO provides a lightweight and plug-and-play solution for improving the visual quality and cross-view consistency of 3D Gaussian representations without additional training or architectural modification.

### *3.2. Anchor-Guided Progressive Optimization*

AGPO uses anchor references as stable local guidance to progressively propagate structural and appearance constraints to neighboring viewpoints. The optimization is organized according to the geometric relationships along the camera trajectory, allowing reliable supervision from sparse anchor viewpoints to gradually extend over the surrounding view space while maintaining coherent geometry and appearance.

Given the input multi-view images and their corresponding camera poses, the selected views serve as anchor references for subsequent optimization. The initial 3DGS representation is rendered from the corresponding anchor viewpoints, and the rendered anchor views are aligned with the input references using a foreground-masked RGB reconstruction loss together with a VGG-based Learned Perceptual Image Patch Similarity (LPIPS) loss. This anchor alignment process improves the fidelity of the initial representation at the anchor viewpoints and establishes reliable references for subsequent propagation.

After anchor alignment, AGPO progressively extends optimization from the anchor regions to neighboring viewpoints. Specifically, the camera trajectory is divided into local regions according to the anchor poses, and propagation proceeds outward from the anchors according to view distance along the trajectory. At propagation distance $k$, all target viewpoints located at the same propagation distance from their corresponding anchor regions are optimized jointly. Before optimizing the targets at distance $k$, the corresponding source views are re-rendered from the current Gaussian representation, and their RGB and depth maps are detached as fixed supervision for the current propagation round. In this way, the optimization advances across the view space in

a deterministic wavefront manner, with all active propagation branches at the same distance contributing to a shared update of the 3D Gaussian representation.

During propagation, anchor-guided constraints are transferred to neighboring viewpoints through depth-guided reprojection. For a pixel $p_r = (u_r, v_r)^\top$ in the reference view $r$, its homogeneous coordinate is denoted as $\tilde{p}_r = (u_r, v_r, 1)^\top$. We adopt the world-to-camera convention $\mathbf{x}_c = \mathbf{R}\mathbf{x}_w + \mathbf{T}$. The corresponding projected position $p_t$ in the target view $t$ is computed as

$$p_t = \pi\left(\mathbf{K}_t\left(\mathbf{R}_t\mathbf{R}_r^{-1}\left(\mathbf{K}_r^{-1}\tilde{p}_r D_r(p_r) - \mathbf{T}_r\right) + \mathbf{T}_t\right)\right), \tag{1}$$

where $\mathbf{K}$, $\mathbf{R}$, and $\mathbf{T}$ denote the camera intrinsic matrix, rotation matrix, and translation vector, respectively. $D_r(p_r)$ denotes the rendered depth at pixel $p_r$ in the reference view, and $\pi(\cdot)$ performs perspective division to obtain the corresponding 2D image coordinate. This reprojection establishes explicit geometric correspondences between adjacent viewpoints and provides cross-view supervision for optimizing the target rendering.

Let $\Omega_k$ denote the set of active reference-target view pairs at propagation distance $k$. The propagation losses from all active branches are aggregated as

$$\mathcal{L}_{\text{prop}}^{(k)} = \frac{1}{|\Omega_k|}\sum_{(r,t)\in\Omega_k}\mathcal{L}_{\text{mv}}^{(r,t)}, \tag{2}$$

where $\mathcal{L}_{\text{mv}}^{(r,t)}$ denotes the reliability-aware multi-view objective between reference view $r$ and target view $t$, which is detailed in Sec. 3.3. The mean aggregation allows all active branches at the same propagation distance to jointly determine a single update of the Gaussian parameters.

To preserve the reliable information established at the anchor viewpoints throughout progressive optimization, we further impose an anchor retention constraint during propagation. The current Gaussian representation is rendered at the anchor poses and

constrained against the corresponding anchor references using foreground-masked reconstruction and VGG-based LPIPS supervision. Denoting this constraint by $\mathcal{L}_{\text{ret}}$, the optimization objective at propagation distance $k$ is

$$\mathcal{L}_{\text{total}}^{(k)} = \mathcal{L}_{\text{prop}}^{(k)} + \lambda_{\text{anchor}} \mathcal{L}_{\text{ret}}, \tag{3}$$

where $\lambda_{\text{anchor}}$ controls the strength of anchor retention. This constraint preserves the fidelity of the anchor viewpoints while progressive supervision is propagated toward neighboring views, providing a stable reference throughout the optimization process.

Through wavefront propagation, geometric reprojection, and anchor retention, AGPO progressively transfers reliable structural and appearance constraints from sparse anchor regions to the surrounding view space, thereby improving both the rendering fidelity and cross-view consistency of the optimized 3D Gaussian representation.

### *3.3. Uncertainty-Aware Reweighting*

Although AGPO establishes explicit geometric correspondences between adjacent viewpoints through anchor-guided reprojection, the resulting cross-view supervision is not uniformly reliable across image regions. Occlusions, depth discontinuities, and inaccurate correspondences may introduce noisy optimization signals in geometrically ambiguous regions. To account for such spatially varying reliability, we introduce UAR, which adaptively modulates pixel-level supervision according to geometric consistency.

For a reference-target view pair $(r, t)$, let $\mathbf{I}_{r\to t}$ and $\mathbf{D}_{r\to t}$ denote the reference RGB image and depth map reprojected to the target viewpoint, respectively. For each valid reprojected pixel $p \in \mathcal{P}_{r,t}$, we define the photometric and depth residuals as

$$\begin{aligned} \delta_{\text{rgb}}^{r,t}(p) &= \|\mathbf{I}_t(p) - \mathbf{I}_{r\to t}(p)\|_2^2, \\ \delta_{\text{d}}^{r,t}(p) &= |\mathbf{D}_t(p) - \mathbf{D}_{r\to t}(p)|, \end{aligned} \tag{4}$$

where $\mathcal{P}_{r,t}$ denotes the set of valid correspondences established by reprojection. The photometric residual measures local appearance disagreement, while the depth residual reflects the geometric consistency of the corresponding pixels.

To further preserve local structures and perceptual appearance across viewpoints, we introduce complementary edge and perceptual constraints:

$$\begin{aligned} \mathcal{L}_{\text{edge}}^{r,t} &= \|\nabla \mathbf{I}_t - \nabla \mathbf{I}_{r\to t}\|_2^2, \\ \mathcal{L}_{\text{perc}}^{r,t} &= d_{\text{LPIPS}}\left(\mathbf{I}_t, \mathbf{I}_{r\to t}\right), \end{aligned} \tag{5}$$

where $\nabla(\cdot)$ denotes the Sobel gradient operator and $d_{\text{LPIPS}}(\cdot,\cdot)$ denotes the LPIPS perceptual distance.

The key idea of UAR is to estimate the reliability of each cross-view correspondence using its geometric consistency. Specifically, the reliability weight associated with pixel $p$ is defined as

$$w^{r,t}(p) = \exp\left(-\frac{\delta_{\text{d}}^{r,t}(p)}{\tau}\right), \tag{6}$$

where $\tau$ controls the sensitivity of the reliability estimate to depth discrepancies. Geometrically consistent correspondences receive weights close to one, whereas correspondences with large depth discrepancies are progressively suppressed.

The resulting reliability weights are incorporated into the photometric reprojection and depth-consistency terms, yielding

$$\mathcal{L}_{\text{reproj}}^{\text{uar},r,t} = \sum_{p\in\mathcal{P}_{r,t}} w^{r,t}(p)\,\delta_{\text{rgb}}^{r,t}(p), \tag{7}$$

and

$$\mathcal{L}_{\text{dc}}^{\text{uar},r,t} = \sum_{p\in\mathcal{P}_{r,t}} w^{r,t}(p)\,\delta_{\text{d}}^{r,t}(p). \tag{8}$$

Finally, the reliability-aware multi-view objective for the reference-target pair $(r,t)$

is formulated as

$$\mathcal{L}_{\text{mv}}^{(r,t)} = \lambda_r \mathcal{L}_{\text{reproj}}^{\text{uar},r,t} + \lambda_e \mathcal{L}_{\text{edge}}^{r,t} + \lambda_p \mathcal{L}_{\text{perc}}^{r,t} + \lambda_{dc} \mathcal{L}_{\text{dc}}^{\text{uar},r,t}, \tag{9}$$

where $\lambda_r$, $\lambda_e$, $\lambda_p$, and $\lambda_{dc}$ balance photometric alignment, structural preservation, perceptual similarity, and depth consistency, respectively. UAR suppresses unreliable correspondences while retaining effective signals in geometrically consistent regions. The resulting objective is aggregated across active view pairs as described in Sec. 3.2.

The overall optimization procedure of AGPO is summarized in Algorithm 1.

**Algorithm 1:** Anchor-Guided Progressive Optimization

**Require:** Initial Gaussians $\Theta_0$, anchors $\{(\mathbf{I}_a, \mathbf{M}_a, \mathbf{T}_a)\}_{a\in\mathcal{A}}$, propagation depth $K$, inner iterations $N$

**Ensure:** Optimized Gaussians $\Theta^*$

$\Theta \leftarrow \Theta_0$; initialize $O_{\text{align}}$
**Stage 1: Anchor Alignment**
**for** $j = 1$ **to** 20 **do**
  **for** each anchor $a \in \mathcal{A}$ **do**
    Render $\widehat{\mathbf{I}}_a$ and compute $\ell_a(\Theta)$
  **end for**
  $\mathcal{L}_{\text{align}} \leftarrow |\mathcal{A}|^{-1} \sum_{a\in\mathcal{A}} \ell_a(\Theta)$; $\Theta \leftarrow \text{AdamStep}(O_{\text{align}}, \Theta, \mathcal{L}_{\text{align}})$
**end for**
Initialize $O_{\text{prop}}$; **Stage 2: Wavefront Propagation**
**for** $k = 1$ **to** $K$ **do**
  Construct $\Omega_k$; re-render and detach source RGB/depth
  **for** $j = 1$ **to** $N$ **do**
    **for** each $(r, t) \in \Omega_k$ **do**
      Render $t$, warp $r$ to $t$, and compute $\mathcal{L}_{\text{mv}}^{(r,t)}$ with UAR
    **end for**
    $\mathcal{L}_{\text{prop}}^{(k)} \leftarrow |\Omega_k|^{-1} \sum_{(r,t)\in\Omega_k} \mathcal{L}_{\text{mv}}^{(r,t)}$; $\mathcal{L}_{\text{ret}} \leftarrow |\mathcal{A}|^{-1} \sum_{a\in\mathcal{A}} \ell_a(\Theta)$
    $\mathcal{L}_{\text{total}}^{(k)} \leftarrow \mathcal{L}_{\text{prop}}^{(k)} + \lambda_{\text{anchor}} \mathcal{L}_{\text{ret}}$; $\Theta \leftarrow \text{AdamStep}(O_{\text{prop}}, \Theta, \mathcal{L}_{\text{total}}^{(k)})$
  **end for**
**end for**
**return** $\Theta^* \leftarrow \Theta$

## 4. Experiments and Results

To comprehensively evaluate the proposed AGPO framework, we conduct extensive experiments under both multi-view 3D generation and reconstruction settings from quantitative and qualitative perspectives. We evaluate whether AGPO improves the rendering fidelity, geometric coherence, and cross-view consistency of the initial 3D Gaussian representations. For the text-conditioned generation setting, we additionally assess the semantic alignment between the rendered views and the input text prompt.

### *4.1. Materials and Methods*

#### *4.1.1. Experimental Setup*

We evaluate AGPO in two settings: text-conditioned 3D generation and multi-view reconstruction. In both settings, AGPO is applied as a training-free post-optimization method: the initialization models remain unchanged, and only the initialized Gaussian representations are optimized using the available multi-view observations and camera geometry.

#### *4.1.2. Dataset*

For the text-to-3D generation setting, we use the $T^3$Bench Single Object benchmark, which contains 100 official object-centric text prompts. These predefined prompts cover diverse semantic categories, appearance attributes, and geometric structures, ranging from relatively simple objects with compact shapes to more challenging objects involving thin structures, articulated components, fine-grained details, or complex topology. The same benchmark setting is used across all compared methods to ensure a consistent evaluation protocol. The reported mean and standard deviation are computed across the corresponding generated objects.

For the multi-view reconstruction setting, we follow the object-level evaluation protocol of FreeSplatter. Specifically, we use Google Scanned Objects (GSO) [39] and OmniObject3D [40], comprising 300 objects across 30 categories. Each object is

represented by 24 views, including four structured input views uniformly distributed at an elevation of 20° and 20 randomly sampled views reserved for novel-view evaluation. Reconstruction quality is assessed using peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and LPIPS. All methods within the same evaluation setting use identical input data and camera configurations. The reported reconstruction statistics are computed across all 300 evaluated objects.

#### *4.1.3. Evaluation Criteria*

We adopt task-specific evaluation metrics for the text-to-3D generation and multi-view reconstruction settings.

For text-to-3D generation, we follow the official evaluation protocol of $T^3$Bench [41]. $T^3$Bench evaluates generated 3D content using a Quality Score and an Alignment Score. The Quality Score assesses the perceptual quality and multi-view consistency of the generated content based on renderings from multiple viewpoints, while the Alignment Score measures the semantic correspondence between the generated 3D representation and the input text through multi-view captioning and language-model-based assessment. Both scores range from 0 to 100, with higher values indicating better performance, and their average defines the $T^3$Bench Overall score. To keep the quantitative comparisons compact, we report the $T^3$Bench Overall score as the primary benchmark metric in the main tables unless otherwise specified.

To further evaluate cross-view consistency, we report multi-view SSIM (MV-SSIM), multi-view LPIPS (MV-LPIPS), and reprojection MSE over neighboring viewpoint pairs. For each reference-target pair, the reference rendering is reprojected to the target view using its rendered depth and the corresponding camera parameters. MV-SSIM and MV-LPIPS respectively measure the structural and perceptual consistency between the reprojected reference and the target rendering, whereas reprojection MSE measures their pixel-level discrepancy over valid overlapping regions. The metrics are averaged over all valid view pairs and evaluation objects. Higher MV-SSIM and lower MV-

LPIPS and reprojection MSE indicate better cross-view consistency. Since ground-truth 3D geometry is generally unavailable in the text-to-3D setting, reprojection MSE is treated as a geometry-related consistency proxy rather than a direct measure of absolute 3D geometric accuracy.

For multi-view reconstruction, we follow standard novel-view synthesis evaluation protocols and report PSNR, SSIM, and LPIPS between rendered novel views and the corresponding ground-truth images. PSNR measures pixel-level reconstruction fidelity, SSIM evaluates structural similarity, and LPIPS measures perceptual discrepancy in deep feature space. Higher PSNR and SSIM values indicate better reconstruction quality, whereas lower LPIPS values indicate better perceptual fidelity. Unless otherwise specified, all reconstruction metrics are computed on the same held-out viewpoints using identical camera configurations for all compared methods.

#### *4.1.4. Implementation Details*

We evaluate AGPO with LGM and FreeSplatter as two 3D Gaussian initialization backbones. In the text-to-3D setting, MVDream generates four canonical views and LGM constructs the initial 3DGS using its native ray-camera configuration with a radius of 1.5 and a vertical field of view of 49.1°. AGPO refinement is performed using the same camera configuration along a 64-view horizontal trajectory at $256 \times 256$ resolution, with the four anchors uniformly located at trajectory indices 0, 16, 32, and 48. In the reconstruction setting, FreeSplatter provides the initial 3DGS together with the camera poses and intrinsics used for anchor alignment and pose-aware propagation. All Gaussian attributes, including position, spherical harmonics (SH) coefficients, opacity, scale, and rotation, are optimized using Adam with $\epsilon = 10^{-15}$. The base learning rates are $1.0 \times 10^{-4}$ for Gaussian positions, $5.0 \times 10^{-3}$ for zeroth-order SH coefficients, $2.5 \times 10^{-4}$ for higher-order SH coefficients, and $5.0 \times 10^{-3}$ for opacity, scale, and rotation. During propagation, these learning rates are scaled by $1/32$ within each distance round.

For both initialization backbones, the four anchor views are jointly optimized using a shared Adam optimizer. At each step, foreground-masked MSE and VGG-based LPIPS losses are computed for all four anchors and averaged, with weights of 2.0 and 0.2, respectively. Anchor alignment is performed for 20 optimization steps. For perceptual supervision, the foreground-composited images are resized to $224 \times 224$ before VGG-based LPIPS evaluation.

Progressive optimization follows a breadth-first wavefront schedule. At each propagation distance, the losses of all active reference–target branches are averaged and combined with the mean fidelity loss of the four anchors. Each distance contains 32 inner iterations, with one shared Adam update per iteration. The weights for reprojection, edge, perceptual, and depth-consistency terms are set to 1.2, 0.10, 0.005, and 0.3, respectively, while UAR uses a depth temperature of $\tau = 0.02$ and the anchor-retention weight is set to 0.18. After each distance round, the optimized target RGB and depth maps are re-rendered and detached as reference signals for the next propagation distance. For the regular 64-view trajectory, the maximum propagation distance is set to $K = 6$, corresponding to six wavefront propagation rounds. When two propagation branches reach the same midpoint target, the view is deterministically assigned to a single branch to avoid duplicate updates. All experiments are conducted on a workstation equipped with two NVIDIA GeForce RTX 3090 GPUs.

### *4.2. Ablation Study*

We conduct comprehensive ablation studies to validate the effectiveness of AGPO and analyze the contributions of its key components and optimization configurations. Specifically, we investigate the overall effectiveness of AGPO, the impact of UAR, the influence of propagation depth and anchor configurations, the contribution of individual loss terms, and the sensitivity of important hyperparameters.

*4.2.1. Component Analysis*

We conduct a component-wise ablation study on the T$^3$Bench Single Object benchmark to evaluate the contribution of each component in AGPO. Starting from the LGM-initialized 3DGS representation, we progressively introduce anchor alignment, progressive propagation, UAR, and anchor retention while keeping the remaining settings unchanged. As shown in Table 1, anchor alignment improves perceptual consistency and reduces reprojection error, while progressive propagation substantially enhances cross-view structural consistency. UAR further suppresses unreliable correspondences and improves the multi-view metrics. The full AGPO model achieves the best T$^3$Bench Overall score, MV-SSIM, and reprojection MSE while maintaining an MV-LPIPS value comparable to that of the UAR-only variant. Compared with the initial 3DGS, Full AGPO improves the T$^3$Bench Overall score by 12.47% and MV-SSIM by 1.72%, while reducing MV-LPIPS and reprojection MSE by 10.65% and 20.78%, respectively.

Table 1: Component-wise ablation study of AGPO. The results demonstrate the contributions of anchor alignment, progressive propagation, UAR, and anchor retention to multi-view consistency.

| **Variant** | **T$^3$Bench Overall** ↑ | **MV-SSIM** ↑ | **MV-LPIPS** ↓ | **Reproj. MSE** ($\times 10^{-3}$) ↓ |
|---|---|---|---|---|
| Initial 3DGS | 46.84 ± 0.76 | 0.9079 ± 0.0032 | 0.0216 ± 0.0005 | 0.929 ± 0.022 |
| + Anchor Alignment | 47.16 ± 0.92 | 0.9079 ± 0.0031 | 0.0205 ± 0.0006 | 0.873 ± 0.024 |
| + Progressive Propagation | 50.37 ± 1.21 | 0.9173 ± 0.0026 | 0.0198 ± 0.0007 | 0.758 ± 0.029 |
| + UAR | 52.21 ± 0.84 | 0.9224 ± 0.0022 | **0.0192 ± 0.0005** | 0.737 ± 0.029 |
| Full AGPO | **52.68 ± 0.64** | **0.9235 ± 0.0025** | 0.0193 ± 0.0006 | **0.736 ± 0.024** |

The comparison between the "+ UAR" variant and Full AGPO isolates the contribution of anchor retention. Adding anchor retention increases the T$^3$Bench Overall score from 52.21 to 52.68 and MV-SSIM from 0.9224 to 0.9235, while slightly reducing the reprojection MSE from $0.737 \times 10^{-3}$ to $0.736 \times 10^{-3}$. MV-LPIPS remains comparable, changing only from 0.0192 to 0.0193. These results indicate that anchor retention provides a modest stabilization effect while preserving comparable perceptual consistency.

The qualitative results in Fig. 2 further illustrate the contribution of each compo-

nent. The initial LGM reconstruction contains noticeable artifacts and incomplete leg structures. Anchor alignment provides a modest improvement, whereas progressive propagation substantially improves the object appearance and restores more coherent structures across viewpoints. UAR further suppresses most of the remaining artifacts, leading to cleaner and more stable renderings. The full AGPO model preserves these improvements and produces the final visually coherent reconstruction.

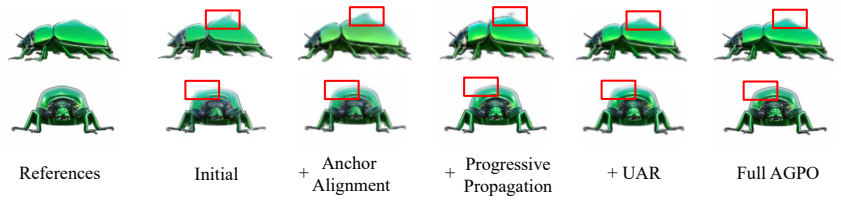



Figure 2: Qualitative component-wise ablation results of AGPO. The first column shows MVDream-generated reference views used to initialize the 3DGS representation and guide subsequent optimization. The initial 3DGS contains noticeable artifacts and incomplete structures. Anchor alignment provides a modest improvement, progressive propagation substantially improves appearance and structural coherence, and UAR suppresses most remaining artifacts. Full AGPO produces clean and consistent renderings.

### *4.2.2. Loss Term Analysis*

We perform a loss-term ablation study on the T$^3$Bench Single Object benchmark by individually removing each loss component from the propagation objective. As shown in Table 2, the full AGPO model achieves the best overall performance, demonstrating that different supervision terms provide complementary constraints for progressive optimization.

Table 2: Ablation study of individual loss terms in the propagation objective of AGPO. The results are reported on the T$^3$Bench Single Object benchmark with mean ± standard deviation over all test objects.

| **Variant** | **T$^3$Bench Overall** ↑ | **MV-SSIM** ↑ | **MV-LPIPS** ↓ | **Reproj. MSE** ($\times 10^{-3}$) ↓ |
|---|---|---|---|---|
| Full AGPO | **52.68 ± 0.64** | **0.9235 ± 0.0025** | 0.0193 ± 0.0006 | **0.736 ± 0.024** |
| w/o $L_{\text{reproj}}$ | 51.74 ± 0.73 | 0.9138 ± 0.0028 | 0.0192 ± 0.0005 | 0.983 ± 0.037 |
| w/o $L_{\text{perc}}$ | 52.21 ± 0.97 | 0.9174 ± 0.0027 | 0.0206 ± 0.0006 | 0.763 ± 0.026 |
| w/o $L_{\text{dc}}$ | 50.95 ± 0.61 | 0.9052 ± 0.0031 | 0.0201 ± 0.0007 | 0.859 ± 0.036 |
| w/o $L_{\text{edge}}$ | 51.31 ± 1.12 | 0.9091 ± 0.0028 | 0.0199 ± 0.0006 | 1.024 ± 0.033 |

Removing the depth consistency term causes the largest degradation in MV-SSIM,

decreasing it from 0.9235 to 0.9052, which verifies the importance of depth-aware geometric guidance for maintaining cross-view structural consistency. The edge consistency term also contributes substantially to geometric coherence, as removing it increases the reprojection MSE from $0.736 \times 10^{-3}$ to $1.024 \times 10^{-3}$. Removing the reprojection term similarly increases the reprojection error and reduces MV-SSIM, despite producing a marginally lower MV-LPIPS value. The removal of the LPIPS term degrades all reported metrics, confirming that perceptual supervision provides complementary appearance guidance. Overall, these results demonstrate that the four loss terms provide complementary constraints during progressive optimization.

### *4.2.3. Propagation Depth Analysis*

We further investigate the effect of propagation depth $K$ on AGPO. As shown in Fig. 3, increasing $K$ initially improves multi-view consistency, since reliable supervision can be propagated to a broader range of neighboring viewpoints. Specifically, MV-SSIM increases steadily from $K = 2$ to $K = 6$, while the reprojection MSE decreases consistently over the same range.

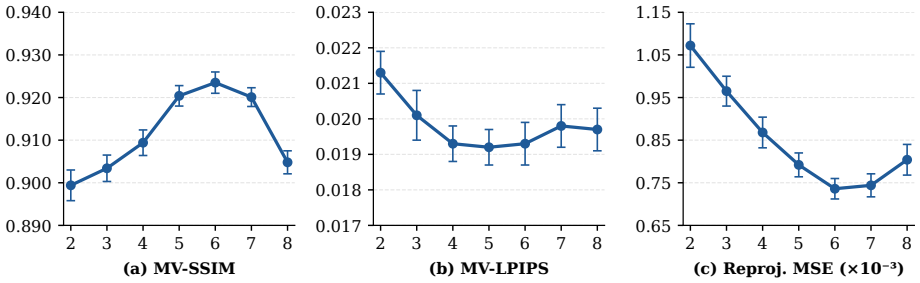



Figure 3: Propagation depth analysis of AGPO. Panels (a)–(c) show MV-SSIM, MV-LPIPS, and reprojection MSE as the propagation depth $K$ varies from 2 to 8, with error bars indicating mean ± standard deviation. Performance reaches the best overall level around $K = 5$–6, while larger depths lead to degraded multi-view consistency.

The quantitative results indicate that the performance reaches a plateau around $K = 5$–6. In particular, $K = 5$ yields the lowest MV-LPIPS, whereas $K = 6$ achieves the highest MV-SSIM and the lowest reprojection MSE with a comparable MV-LPIPS

value. Since the numerical differences between $K = 5$ and $K = 6$ are small, we do not interpret them as evidence that either setting is statistically superior. When the propagation depth is increased beyond $K = 6$, the multi-view consistency metrics begin to deteriorate, suggesting that overly long propagation may accumulate unreliable cross-view supervision. Meanwhile, the runtime increases monotonically with $K$. Accordingly, we adopt $K = 6$ as the default configuration because it provides consistently strong performance across the three multi-view metrics while remaining at the boundary before further propagation begins to cause degradation.

#### *4.2.4. Anchor Configuration Analysis*

We investigate the effect of anchor configuration by varying the number of anchor views while keeping all other settings unchanged. As shown in Table 3, increasing the number of anchors improves multi-view consistency and reprojection consistency. Using only one anchor view results in lower performance because of its limited supervision coverage. Increasing the number of anchors from one to two and four progressively improves the overall results. The four-anchor configuration achieves the best overall performance, obtaining an MV-SSIM of 0.9235 and the lowest reprojection MSE of $0.736 \times 10^{-3}$. Therefore, we adopt four uniformly distributed anchors as the default configuration of AGPO.

Table 3: Anchor configuration analysis of AGPO. The results are reported on the $T^3$Bench Single Object benchmark with mean ± standard deviation over all test objects.

| Configuration | MV-SSIM ↑ | MV-LPIPS ↓ | Reproj. MSE ($\times 10^{-3}$) ↓ |
|---|---|---|---|
| 1 Anchor | 0.8915 ± 0.0029 | 0.0211 ± 0.0010 | 1.088 ± 0.059 |
| 2 Anchors | 0.9031 ± 0.0038 | 0.0208 ± 0.0006 | 0.963 ± 0.031 |
| 4 Anchors | **0.9235 ± 0.0025** | **0.0193 ± 0.0006** | **0.736 ± 0.024** |

#### *4.2.5. Hyperparameter Sensitivity*

We investigate the sensitivity of the anchor retention weight $\lambda_{\text{anchor}}$ and the UAR temperature $\tau$ using one-factor-at-a-time sweeps around their default values. As shown

in Fig. 4, AGPO maintains relatively stable performance under moderate variations of either hyperparameter.

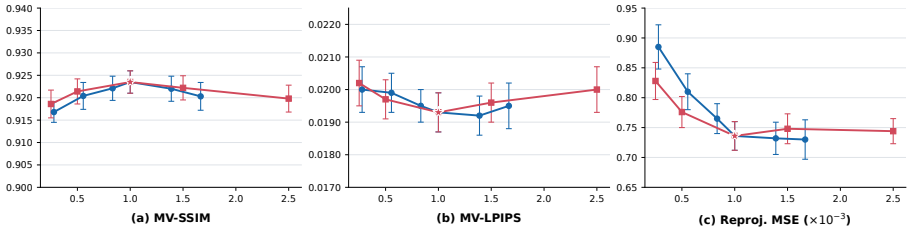



Figure 4: **Hyperparameter sensitivity analysis.** Blue circles and red squares denote one-factor-at-a-time sweeps of the anchor retention weight $\lambda_{\text{anchor}}$ and the UAR temperature $\tau$, respectively. The horizontal coordinate reports each hyperparameter relative to its default value, $x/x_0$, where $\lambda_{\text{anchor},0} = 0.18$ and $\tau_0 = 0.020$. The star marks the default configuration at $x/x_0 = 1$. Error bars indicate mean ± standard deviation over all evaluated objects. Higher MV-SSIM is better, whereas lower MV-LPIPS and reprojection MSE are better.

For $\lambda_{\text{anchor}}$, increasing the weight from a relatively small value improves multi-view consistency by strengthening the preservation of reliable anchor supervision. However, an excessively large value over-constrains the anchor views and reduces the flexibility of progressive propagation. The default setting therefore provides a suitable balance between anchor fidelity and propagation flexibility.

The UAR temperature $\tau$ controls the rate at which correspondence weights decrease with increasing depth discrepancy. A very small temperature may over-suppress potentially useful supervision, whereas a large temperature may retain geometrically unreliable correspondences. The results show that the default value provides a favorable balance across MV-SSIM, MV-LPIPS, and reprojection MSE. Therefore, we use $\lambda_{\text{anchor}} = 0.18$ and $\tau = 0.02$ as the default settings.

### *4.3. Quantitative Comparisons*

#### *4.3.1. Text-to-3D Generation Enhancement*

To evaluate AGPO as a 3D Gaussian representation enhancement framework, we compare it with representative enhancement methods, including SuperGaussian and 3DEnhancer. Unlike conventional text-to-3D methods that generate 3D assets from

scratch, these approaches refine existing 3D representations. For a fair comparison, all methods are applied to the same LGM-initialized 3D Gaussian representations and evaluated on the T$^3$Bench Single Object benchmark using T$^3$Bench Overall, MV-SSIM, MV-LPIPS, and reprojection MSE.

Table 4: Quantitative comparison with different 3D representation enhancement methods on the T$^3$Bench Single Object benchmark. Results are reported as mean ± standard deviation over all test objects.

| **Method** | **T$^3$Bench Overall** ↑ | **MV-SSIM** ↑ | **MV-LPIPS** ↓ | **Reproj. MSE** ($\times 10^{-3}$) ↓ |
|---|---|---|---|---|
| Initial LGM | 46.84 ± 0.76 | 0.9079 ± 0.0032 | 0.0216 ± 0.0005 | 0.929 ± 0.022 |
| SuperGaussian | 49.36 ± 1.18 | 0.9148 ± 0.0029 | 0.0204 ± 0.0009 | 0.842 ± 0.025 |
| 3DEnhancer | 46.27 ± 1.05 | 0.9128 ± 0.0026 | 0.0198 ± 0.0006 | 0.981 ± 0.043 |
| **AGPO (Ours)** | **52.68 ± 0.64** | **0.9235 ± 0.0025** | **0.0193 ± 0.0006** | **0.736 ± 0.024** |

As shown in Table 4, SuperGaussian and 3DEnhancer improve selected aspects of the initial LGM representations, demonstrating the effectiveness of post-generation enhancement. However, their improvements in appearance do not consistently translate into better cross-view reprojection consistency. In contrast, AGPO achieves the best performance across all metrics, increasing the T$^3$Bench Overall score from 46.84 to 52.68 and reducing the reprojection MSE from $0.929 \times 10^{-3}$ to $0.736 \times 10^{-3}$. These results demonstrate the effectiveness of organizing multi-view supervision through anchor-guided propagation and uncertainty-aware reweighting.

To assess statistical significance, we conduct two-sided paired Wilcoxon signed-rank tests on the per-sample results of Initial LGM and AGPO under the same evaluation setting. Table 5 reports the paired differences and corresponding $p$-values. The improvements are statistically significant across all evaluated metrics.

Table 5: Two-sided paired Wilcoxon signed-rank tests between Initial LGM and AGPO. Differences are computed as AGPO minus Initial LGM.

| **Metric** | **Mean Diff.** | **Median Diff.** | $p$**-value** |
|---|---|---|---|
| T$^3$Bench Overall ↑ | +5.84 | +5.63 | 0.0002 |
| MV-SSIM ↑ | +0.0156 | +0.0128 | 0.0001 |
| MV-LPIPS ↓ | -0.0023 | -0.0020 | 0.0028 |
| Reprojection MSE ($\times 10^{-3}$) ↓ | -0.193 | -0.156 | $< 0.001$ |

*4.3.2. Generalization to Reconstruction-based Initialization*

To further verify the generalization ability of AGPO beyond text-to-3D generation, we apply AGPO to 3D Gaussian representations initialized by FreeSplatter, a pose-free sparse-view reconstruction method. Unlike LGM, which constructs 3D Gaussian representations from diffusion-generated multi-view observations, FreeSplatter initializes 3D Gaussians from sparse input images while estimating the corresponding camera parameters. In this experiment, we directly optimize the FreeSplatter-initialized Gaussian representations with AGPO, while keeping the original reconstruction process unchanged.

As shown in Table 6, AGPO consistently improves the novel-view synthesis quality of FreeSplatter-initialized representations across all evaluation metrics. Specifically, AGPO improves PSNR from 26.36 to 29.18 dB and increases SSIM from 0.8821 to 0.9064, while reducing LPIPS from 0.1427 to 0.1283. These improvements demonstrate that AGPO can effectively refine Gaussian representations produced by sparse-view reconstruction pipelines. This result further indicates that AGPO is not tied to a single 3D Gaussian initialization strategy. Instead, it serves as a general post-optimization strategy that exploits initialized 3D Gaussian representations, associated camera information, and available multi-view observations to improve rendering fidelity and cross-view consistency across different initialization strategies.

Table 6: Quantitative comparison of AGPO on FreeSplatter-initialized 3D Gaussian representations. The results are reported with mean ± standard deviation over all evaluated objects.

| **Method** | **PSNR ↑** | **SSIM ↑** | **LPIPS ↓** |
|---|---|---|---|
| FreeSplatter | 26.36 ± 1.65 | 0.8821 ± 0.0315 | 0.1427 ± 0.0186 |
| FreeSplatter + AGPO | **29.18 ± 1.48** | **0.9064 ± 0.0289** | **0.1283 ± 0.0172** |

### *4.4. Qualitative Comparisons*

#### *4.4.1. Text-to-3D Generation Enhancement*

To further demonstrate the effectiveness of AGPO in enhancing 3D Gaussian representations, we provide qualitative comparisons with existing Gaussian refinement methods in Fig. 5. All methods start from the same LGM-initialized 3D Gaussian representations and perform subsequent optimization for quality enhancement. Although existing approaches improve the initialized representations to some extent, they may still suffer from limitations in appearance preservation, texture fidelity, and cross-view consistency.

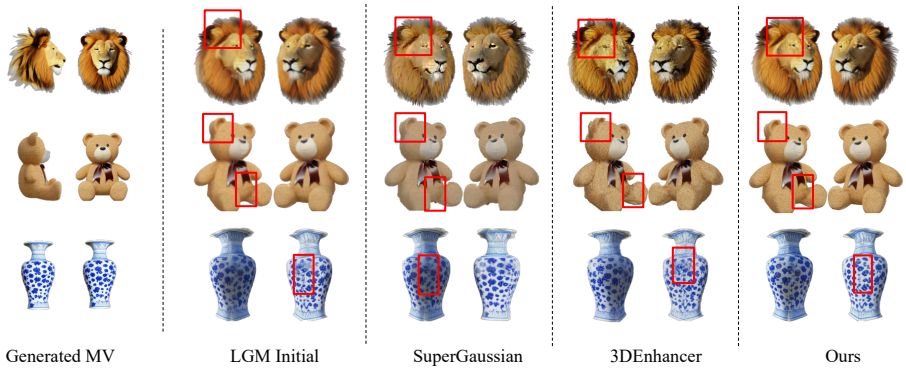



Figure 5: Qualitative comparison of Gaussian refinement methods. MVDream-generated multi-view images are first used by LGM to initialize the 3D Gaussian representation. Based on the same initialization, SuperGaussian, 3DEnhancer, and AGPO (Ours) perform subsequent Gaussian optimization. AGPO produces more consistent and detailed results across different viewpoints.

Specifically, SuperGaussian tends to produce results with reduced color fidelity and relatively grayish appearances after enhancement. While 3DEnhancer improves texture richness and introduces more appearance variations, it tends to smooth the reconstructed representations, leading to blurred fine details and reduced surface sharpness. In contrast, AGPO effectively preserves fine-grained appearance details while maintaining structural completeness and multi-view consistency. As shown in Fig. 5, AGPO produces clearer surface textures, sharper local details, and more consistent renderings across different viewpoints, especially in regions with complex boundaries and

fine-grained appearances. These results demonstrate that AGPO effectively enhances existing 3D Gaussian representations by exploiting geometric relationships and cross-view consistency constraints, without requiring retraining of the underlying generation model.

#### *4.4.2. Reconstruction-based Initialization*

To evaluate the generalization of AGPO beyond text-to-3D generation, we apply it to 3D Gaussian representations initialized by FreeSplatter. Unlike the LGM-based setting, in which the initial representation is constructed from diffusion-generated multi-view observations, FreeSplatter reconstructs 3D Gaussians from sparse input images while estimating their camera parameters. We consider two observation layouts: four regularly distributed horizontal views following the MVDream camera configuration and four views with irregular camera poses.

As shown in Fig. 6, FreeSplatter recovers the overall object structure under both camera configurations, but its renderings contain blurred textures and incomplete local details. AGPO produces clearer structures and textures while maintaining more consistent appearances across the rendered viewpoints. The improvements obtained with both regular and irregular camera layouts indicate that AGPO is not restricted to uniformly distributed input viewpoints.

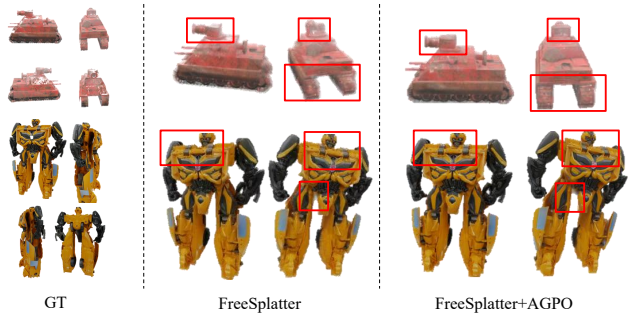



Figure 6: Qualitative results on FreeSplatter-initialized 3D Gaussian representations. The top example uses four regularly distributed horizontal input views following the MVDream camera configuration, whereas the bottom example uses four input views with irregular camera poses. From left to right, we show the ground-truth input views used by FreeSplatter, the initial FreeSplatter reconstruction, and the AGPO-refined result. AGPO improves local structure and texture quality under both camera configurations.

### *4.5. User Study*

To further evaluate the perceptual improvements introduced by AGPO, we conduct a user study comparing the LGM-initialized 3D Gaussian representations with the corresponding results refined by AGPO. We randomly select 50 objects from the $T^3$Bench Single Object benchmark and render each representation from multiple viewpoints. A total of 50 participants are invited to evaluate the visual quality of the generated results.

For each comparison, participants are presented with two results generated from the same input prompt and asked to select the preferred result separately according to four criteria: (1) **geometry quality**, which evaluates the completeness and structural correctness of the generated object; (2) **appearance quality**, which assesses visual fidelity and surface details; (3) **multi-view consistency**, which measures the stability of object geometry and appearance across different viewpoints; and (4) **overall preference**, which reflects the participant's comprehensive assessment of the two results. The preference rate for each method is calculated as the percentage of selections received under the corresponding criterion. Therefore, the preference rates of the two methods sum to 100% for each criterion.

The results are summarized in Table 7. Compared with the LGM-initialized representations, the results refined by AGPO receive higher preference rates across all evaluation criteria. In particular, the higher preference for geometry quality and multi-view consistency indicates that AGPO improves structural completeness and produces more stable renderings across viewpoints. These results demonstrate that the improvements achieved by AGPO are not limited to quantitative metrics but are also perceptually meaningful to human observers.

Table 7: User study results comparing the LGM-initialized representations and AGPO-enhanced results. The preference rates are reported as percentages, and the two methods sum to 100% for each criterion.

| Method | Geometry Quality ↑ | Appearance Quality ↑ | Multi-view Consistency ↑ | Overall Preference ↑ |
|---|---|---|---|---|
| LGM | 46.0 | 48.9 | 47.3 | 46.1 |
| LGM + AGPO | **54.0** | **51.1** | **52.7** | **53.9** |

*4.6. Discussion*

The experimental results demonstrate that AGPO provides an effective training-free post-optimization strategy for enhancing existing 3D Gaussian representations. By progressively propagating structural and appearance constraints from observed viewpoints while suppressing unreliable correspondences through UAR, AGPO improves rendering quality and cross-view consistency without introducing additional generative priors.

Nevertheless, AGPO has two main limitations. First, its performance depends on the quality of the initialized 3D Gaussian representation and the reliability of the available multi-view observations. Severe geometric errors or missing structures may produce inaccurate depth estimates and cross-view correspondences that persist or are further propagated during optimization. Semantic drift or structural inconsistency among anchor views may likewise introduce biased supervision. Without external geometric priors or an independent verification mechanism, AGPO may therefore be unable to recover substantially incorrect or missing object details, as illustrated in Fig. 7.

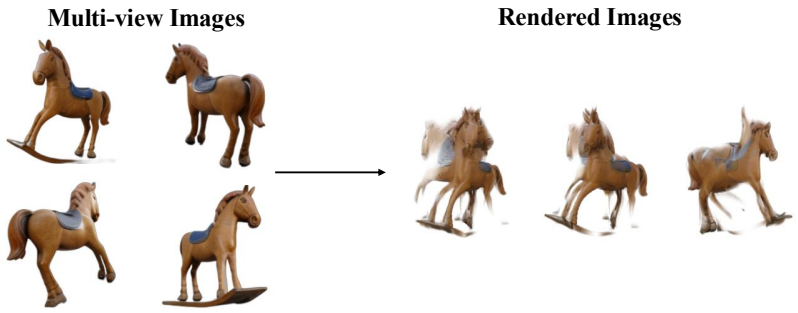



Figure 7: Failure case under inconsistent multi-view observations. Structural inconsistencies and missing content in the input views (left) are inherited by the reconstructed 3D Gaussian representation, producing severe geometric artifacts in the rendered views (right) that AGPO cannot fully correct during post-optimization.

Second, although AGPO has been evaluated with different viewpoint configurations and anchor settings, the current framework remains primarily designed for object-centric representations with sufficiently accurate camera poses, adequate view overlap,

and reliable local correspondences. Extending it to large-scale scenes is challenging because non-uniform viewpoint coverage, substantial depth variation, severe occlusion, and longer propagation paths may increase correspondence uncertainty and error accumulation. Future work will investigate cross-anchor validation, external geometric guidance, and visibility-aware propagation over scene-adaptive view graphs.

## 5. Conclusion

We presented AGPO, a lightweight and training-free framework for enhancing initialized 3D Gaussian representations. AGPO organizes available multi-view observations as anchor supervision and progressively transfers structural and appearance constraints across the view space through depth-guided reprojection and wavefront optimization. Anchor retention preserves fidelity to directly observed views, while UAR reduces the influence of unreliable correspondences caused by occlusions and inaccurate geometry. Experiments on text-conditioned 3D generation and multi-view reconstruction demonstrate consistent improvements in visual fidelity and cross-view consistency across different initialization strategies. Ablation studies further validate the contributions of the proposed components. Overall, AGPO provides a plug-and-play refinement strategy without retraining or modifying the underlying generation or reconstruction models.

## CRediT authorship contribution statement

**Jiaheng Xu**: Writing – original draft, Writing – review & editing, Data curation, Software; **Xuran Zhao**, **Tao Yang**, **Yunuo Zou**: Writing – review & editing; **Yan Tian**: Supervision, Conceptualization, Methodology, Resources, Funding acquisition, Writing – review & editing; **Rui Zhang**: Methodology, Resources, Writing – review & editing; **Xun Wang**: Funding acquisition, Writing – review & editing.

## Declaration of Competing Interest

All authors declare that they have no conflicts of interest with respect to the publication of this article.

## Availability of Data and Materials

Data are available from the corresponding author upon reasonable request.

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